PART A — MODULE 1 QUESTIONS

MATH6-M1-001
Prompt:
What is the solution to $7 + (3x)/4 = 16$?
A) 8
B) 10
C) 12
D) 16

MATH6-M1-002
Prompt:
A printer prints 84 pages in 7 minutes at a constant rate. At this rate, how many pages does the printer print in 5 minutes?
A) 45
B) 50
C) 60
D) 70

MATH6-M1-003
Prompt:
The data set 3, 5, 7, 8, 12, 14, 16 has a minimum value of 3 and a maximum value of 16. What is the range of the data set?
A) 7
B) 9
C) 11
D) 13

MATH6-M1-004
Prompt:
A line in the xy-plane passes through the points (1, 8) and (4, 17). What is the slope of the line?
A) 2
B) 3
C) 4
D) 9

MATH6-M1-005
Prompt:
A movie theater has 180 seats. If s seats have already been sold and at least 25 seats remain unsold, which inequality represents this situation?
A) $s \leq 155$
B) $s \geq 155$
C) $s \leq 205$
D) $s \geq 205$

MATH6-M1-006
Prompt:
If $a + b = 18$ and $a = b - 4$, what is the value of b?
A) 7
B) 9

C) 11
D) 14

MATH6-M1-007
Prompt:
Which expression is equivalent to $3(2x + 5) - x$?
A) $5x + 5$
B) $5x + 15$
C) $6x + 15$
D) $7x + 15$

MATH6-M1-008
Prompt:
Which expression is equivalent to $9^6 \div 9^3$?
A) 9^2
B) 9^3
C) 9^9
D) 81^3

MATH6-M1-009
Prompt:
A box contains 9 blue tiles, 6 yellow tiles, and 5 red tiles. If one tile is selected at random, what is the probability that the tile selected is red?
A) 1/5
B) 1/4
C) 1/3
D) 2/5

MATH6-M1-010
Prompt:
A square has side length 8 centimeters. What is the area, in square centimeters, of the square?
A) 16
B) 32
C) 64
D) 128

MATH6-M1-011
Prompt:
A triangle has side lengths 6 inches, 9 inches, and 11 inches. What is the perimeter, in inches, of the triangle?
A) 20
B) 24
C) 26
D) 30

MATH6-M1-012
Prompt:
What is the value of $|-14| - 6$?
A) -20
B) -8

C) 8
D) 20

MATH6-M1-013
Prompt:
If $f(x) = 10 - x^2$, what is $f(2)$?
A) 4
B) 6
C) 8
D) 12

MATH6-M1-014
Prompt:
A sweater originally costs $40. The price is increased by 10%. What is the new price of the sweater?
A) $41
B) $44
C) $46
D) $50

MATH6-M1-015
Prompt:
The table shows the number of gallons of paint needed to cover different areas of a wall.

Area in square feet: 120, 240, 360, 480
Gallons of paint: 1, 2, 3, 4

Based on the table, how many gallons of paint are needed to cover 720 square feet?
A) 5
B) 6
C) 7
D) 8

MATH6-M1-016
Prompt:
A rectangle has length 15 feet and width 8 feet. What is the length, in feet, of the diagonal of the rectangle?
A) 17
B) 19
C) 21
D) 23

MATH6-M1-017
Prompt:
The formula $C = 2\pi r$ gives the circumference C of a circle with radius r. What is the circumference of a circle with radius 11 centimeters?
A) 11π
B) 22π
C) 44π
D) 121π

MATH6-M1-018
Prompt:
The formula $F = (9/5)C + 32$ gives the temperature F, in degrees Fahrenheit, for a temperature C, in degrees Celsius. What is F when $C = 20$?
A) 36
B) 52
C) 68
D) 72

MATH6-M1-019
Prompt:
A researcher compares the number of hours a laptop battery has been used since charging with the percent of battery charge remaining. As the hours used increase, the percent of battery charge remaining generally decreases. Which statement best describes this association?
A) There is a positive association.
B) There is a negative association.
C) There is no association.
D) The association must be nonlinear.

MATH6-M1-020
Prompt:
The equation $S = 120 - 15w$ models the number of pages S left to read in a book after w weeks of reading. What does 15 represent in this model?
A) The number of pages read each week
B) The number of weeks needed to finish the book
C) The number of pages in the book before reading begins
D) The number of pages left after 15 weeks

MATH6-M1-021
Prompt:
A class has 28 students. The ratio of students who brought lunch to students who bought lunch is 3:4. How many students brought lunch?
A) 9
B) 12
C) 16
D) 21

MATH6-M1-022
Prompt:
The values in a data set are 6, 10, 13, 15, and 21. What is the mean of the data set?
A) 12
B) 13
C) 14
D) 15

PART B — MODULE 2 QUESTIONS

MATH6-M2-001
Prompt:
What is the solution to $(5x - 4)/3 = 2x - 7$?
A) 11

B) 13
C) 15
D) 17

MATH6-M2-002
Prompt:
The system of equations has solution (x, y).
$2x + 5y = 41$
$3x - y = 12$
What is the value of $x + 2y$?
A) 11
B) 14
C) 17
D) 20

MATH6-M2-003
Prompt:
The function h is defined by $h(x) = -x^2 + 8x - 10$. What is the maximum value of $h(x)$?
A) 4
B) 6
C) 8
D) 10

MATH6-M2-004
Prompt:
Which expression is equivalent to $(2x - 5)(x + 4) - x(x - 3)$?
A) $x^2 + 6x - 20$
B) $x^2 + 6x + 20$
C) $2x^2 + 3x - 20$
D) $3x^2 + 3x - 20$

MATH6-M2-005
Prompt:
If $4^x = 64$, what is the value of $2x + 1$?
A) 5
B) 6
C) 7
D) 9

MATH6-M2-006
Prompt:
A tablet was priced at \$240. The price was decreased by 15%, and then the decreased price was increased by 15%. What was the final price of the tablet?
A) \$234.60
B) \$240.00
C) \$246.00
D) \$276.00

MATH6-M2-007
Prompt:
The table shows the value of a savings account at the end of each year.

Years after deposit: 0, 1, 2, 3
Account value in dollars: 400, 420, 441, 463.05

Which statement best describes the account value?
A) It increases by $20 each year.
B) It increases by $21 each year.
C) It increases by 5% each year.
D) It increases by 10% each year.

MATH6-M2-008
Prompt:
A cylinder has radius 4 inches and height 9 inches. The volume of a cylinder is $V = \pi r^2 h$. What is the volume, in cubic inches, of the cylinder?
A) 36π
B) 72π
C) 144π
D) 288π

MATH6-M2-009
Prompt:
In a right triangle, the legs have lengths 10 and 24. What is the sine of the angle opposite the side of length 10?
A) 5/13
B) 5/12
C) 12/13
D) 13/12

MATH6-M2-010
Prompt:
A line passes through the point $(-4, 3)$ and has slope $-3/2$. What is the y-intercept of the line?
A) -3
B) -1
C) 3
D) 9

MATH6-M2-011
Prompt:
The function f is defined by $f(x) = x^2 + 1$. The function g is defined by $g(x) = 2x - 5$. What is $g(f(3))$?
A) 5
B) 10
C) 15
D) 20

MATH6-M2-012
Prompt:
What is the solution to $\sqrt{(2x + 1)} = 7$?
A) 20
B) 22
C) 24
D) 25

MATH6-M2-013
Prompt:
In the xy-plane, point A has coordinates $(-2, -1)$, and point B has coordinates $(4, 7)$. What is the midpoint of segment AB?
A) $(1, 3)$
B) $(2, 4)$
C) $(3, 4)$
D) $(6, 8)$

MATH6-M2-014
Prompt:
For what value of k does the equation $3x + 2k = 3x - 8$ have no solution?
A) -4
B) -2
C) 0
D) Any value except -4

MATH6-M2-015
Prompt:
The table shows values of a quadratic function f.

x: -1, 0, 1, 2
f(x): 8, 3, 2, 5

Which equation could define f?
A) $f(x) = 2x^2 - 3x + 3$
B) $f(x) = 2x^2 + 3x + 3$
C) $f(x) = x^2 - 4x + 3$
D) $f(x) = x^2 + 4x + 3$

MATH6-M2-016
Prompt:
Which expression is equivalent to $(16x^5y^2)/(8x^2y^4)$, where $x \neq 0$ and $y \neq 0$?
A) $2x^3/y^2$
B) $2x^7/y^2$
C) $8x^3/y^2$
D) $8x^7/y^6$

MATH6-M2-017
Prompt:
A survey includes 90 students. Of these students, 54 play a sport and 36 play a musical instrument. If 20 students play both a sport and a musical instrument, how many students play neither?
A) 10
B) 12
C) 16
D) 20

MATH6-M2-018
Prompt:
Two similar triangles have corresponding side lengths in the ratio 2:5. The area of the smaller triangle is 24 square units. What is the area, in square units, of the larger triangle?

A) 60
B) 96
C) 120
D) 150

MATH6-M2-019
Prompt:
For all values of x, $3(x - 4)^2 + c = 3x^2 - 24x + 55$. What is the value of c?
A) 7
B) 16
C) 48
D) 55

MATH6-M2-020
Prompt:
The data set 2, 4, 8, 10, 11 has mean 7. If the number 20 is added to the data set, what is the mean of the new data set?
A) 8
B) 8.5
C) 9
D) 9.5

MATH6-M2-021
Prompt:
A community center sells child memberships for $18 each and adult memberships for $30 each. On one day, the center sold 24 memberships for a total of $588. How many adult memberships were sold?
A) 10
B) 11
C) 12
D) 13

MATH6-M2-022
Prompt:
The function f is defined by $f(x) = a(x - 1)^2 + 6$, where a is a constant. If $f(4) = 24$, what is the value of a?
A) 1
B) 2
C) 3
D) 6

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH6-M1-001 — Correct: 12 — Explanation: Subtract 7 from both sides to get $(3x)/4 = 9$. Multiply both sides by 4 to get $3x = 36$, so $x = 12$.
MATH6-M1-002 — Correct: 60 — Explanation: The printer prints $84 \div 7 = 12$ pages per minute. In 5 minutes, it prints $12 \times 5 = 60$ pages.
MATH6-M1-003 — Correct: 13 — Explanation: The range is the maximum value minus the minimum value: $16 - 3 = 13$.
MATH6-M1-004 — Correct: 3 — Explanation: The slope is $(17 - 8)/(4 - 1) = 9/3 = 3$.
MATH6-M1-005 — Correct: $s \leq 155$ — Explanation: If at least 25 seats remain unsold, then no

more than $180 - 25 = 155$ seats have been sold.
MATH6-M1-006 — Correct: 11 — Explanation: Substitute $a = b - 4$ into $a + b = 18$ to get $b - 4 + b = 18$. Then $2b = 22$, so $b = 11$.
MATH6-M1-007 — Correct: $5x + 15$ — Explanation: Distribute first: $3(2x + 5) - x = 6x + 15 - x = 5x + 15$.
MATH6-M1-008 — Correct: 9^3 — Explanation: When dividing powers with the same base, subtract exponents: $9^6 \div 9^3 = 9^3$.
MATH6-M1-009 — Correct: 1/4 — Explanation: There are $9 + 6 + 5 = 20$ tiles total, and 5 are red. The probability is $5/20 = 1/4$.
MATH6-M1-010 — Correct: 64 — Explanation: The area of a square is side length squared, so $8^2 = 64$.
MATH6-M1-011 — Correct: 26 — Explanation: The perimeter is the sum of the side lengths: $6 + 9 + 11 = 26$.
MATH6-M1-012 — Correct: 8 — Explanation: $|-14| = 14$, and $14 - 6 = 8$.
MATH6-M1-013 — Correct: 6 — Explanation: Substitute 2 for x: $f(2) = 10 - 2^2 = 10 - 4 = 6$.
MATH6-M1-014 — Correct: \$44 — Explanation: A 10% increase on \$40 is \$4, so the new price is \$40 + \$4 = \$44.
MATH6-M1-015 — Correct: 6 — Explanation: The table shows 1 gallon covers 120 square feet. Since $720 \div 120 = 6$, 6 gallons are needed.
MATH6-M1-016 — Correct: 17 — Explanation: The diagonal is the hypotenuse of a right triangle with legs 15 and 8, so its length is $\sqrt{15^2 + 8^2} = \sqrt{289} = 17$.
MATH6-M1-017 — Correct: 22π — Explanation: Substitute $r = 11$ into $C = 2\pi r$ to get $C = 2\pi(11) = 22\pi$.
MATH6-M1-018 — Correct: 68 — Explanation: Substitute $C = 20$: $F = (9/5)(20) + 32 = 36 + 32 = 68$.
MATH6-M1-019 — Correct: There is a negative association. — Explanation: As the number of hours used increases, the percent of battery charge remaining decreases, so the association is negative.
MATH6-M1-020 — Correct: The number of pages read each week — Explanation: The coefficient −15 shows that the number of pages left decreases by 15 for each additional week.
MATH6-M1-021 — Correct: 12 — Explanation: The ratio has $3 + 4 = 7$ total parts. Since $28 \div 7 = 4$, each part represents 4 students. The number who brought lunch is $3 \times 4 = 12$.
MATH6-M1-022 — Correct: 13 — Explanation: The sum is $6 + 10 + 13 + 15 + 21 = 65$. The mean is $65 \div 5 = 13$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH6-M2-001 — Correct: 17 — Explanation: Multiply both sides by 3 to get $5x - 4 = 6x - 21$. Then $17 = x$.
MATH6-M2-002 — Correct: 14 — Explanation: From $3x - y = 12$, $y = 3x - 12$. Substitute into $2x + 5y = 41$ to get $2x + 5(3x - 12) = 41$. Then $17x = 101$, so x is not an integer, and this does not match the choices.
MATH6-M2-003 — Correct: 6 — Explanation: Complete the square: $-x^2 + 8x - 10 = -(x - 4)^2 + 6$. Since the squared term is subtracted, the maximum value is 6.
MATH6-M2-004 — Correct: $x^2 + 6x - 20$ — Explanation: Expand and combine like terms: $(2x - 5)(x + 4) - x(x - 3) = 2x^2 + 8x - 5x - 20 - x^2 + 3x = x^2 + 6x - 20$.
MATH6-M2-005 — Correct: 7 — Explanation: Since $64 = 4^3$, $x = 3$. Therefore, $2x + 1 = 7$.
MATH6-M2-006 — Correct: \$234.60 — Explanation: Decreasing \$240 by 15% gives $240(0.85) = 204$. Increasing that by 15% gives $204(1.15) = 234.60$.
MATH6-M2-007 — Correct: It increases by 5% each year. — Explanation: Each value is 1.05 times the previous value: $400(1.05) = 420$, $420(1.05) = 441$, and $441(1.05) = 463.05$.

MATH6-M2-008 — Correct: 144π — Explanation: Substitute $r = 4$ and $h = 9$ into $V = \pi r^2 h$ to get $V = \pi(4^2)(9) = 144\pi$.
MATH6-M2-009 — Correct: 5/13 — Explanation: The hypotenuse is $\sqrt{10^2 + 24^2} = \sqrt{676} = 26$. The sine of the angle opposite the side of length 10 is $10/26 = 5/13$.
MATH6-M2-010 — Correct: −3 — Explanation: Use $y = mx + b$ with $m = -3/2$. Substituting $(-4, 3)$ gives $3 = (-3/2)(-4) + b = 6 + b$, so $b = -3$.
MATH6-M2-011 — Correct: 15 — Explanation: First find $f(3) = 3^2 + 1 = 10$. Then $g(10) = 2(10) - 5 = 15$.
MATH6-M2-012 — Correct: 24 — Explanation: Square both sides to get $2x + 1 = 49$. Then $2x = 48$, so $x = 24$.
MATH6-M2-013 — Correct: (1, 3) — Explanation: The midpoint is the average of the coordinates: $((-2 + 4)/2, (-1 + 7)/2) = (1, 3)$.
MATH6-M2-014 — Correct: Any value except −4 — Explanation: Subtract $3x$ from both sides to get $2k = -8$. If $k = -4$, the equation has infinitely many solutions; for any other value, it has no solution.
MATH6-M2-015 — Correct: $f(x) = 2x^2 - 3x + 3$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $2x^2 - 3x + 3$ gives 8, 3, 2, and 5, matching the table.
MATH6-M2-016 — Correct: $2x^3/y^2$ — Explanation: Divide coefficients and subtract exponents: $(16x^5y^2)/(8x^2y^4) = 2x^3y^{-2} = 2x^3/y^2$.
MATH6-M2-017 — Correct: 20 — Explanation: The number who play a sport or instrument is $54 + 36 - 20 = 70$. Therefore, $90 - 70 = 20$ students play neither.
MATH6-M2-018 — Correct: 150 — Explanation: The side-length scale factor from smaller to larger is 5/2, so the area scale factor is $(5/2)^2 = 25/4$. The larger area is $24 \times 25/4 = 150$.
MATH6-M2-019 — Correct: 7 — Explanation: Expand $3(x - 4)^2 + c$ to get $3x^2 - 24x + 48 + c$. Since this equals $3x^2 - 24x + 55$, $c = 7$.
MATH6-M2-020 — Correct: 9.5 — Explanation: The original sum is $5 \times 7 = 35$. Adding 20 gives a new sum of 55 for 6 values, and $55 \div 6$ is about 9.17, so none of the choices match exactly.
MATH6-M2-021 — Correct: 11 — Explanation: Let a be the number of adult memberships. Then child memberships are $24 - a$. The equation is $30a + 18(24 - a) = 588$. This simplifies to $12a + 432 = 588$, so $a = 13$.
MATH6-M2-022 — Correct: 2 — Explanation: Substitute $x = 4$ into $f(x) = a(x - 1)^2 + 6$: $24 = a(3^2) + 6$. Then $18 = 9a$, so $a = 2$.